The Golden Layer: A Hyperbolic Construction in Swirl–String Theory

Unveiling a distinguished kinematic state defined by the golden ratio.

Based on the work of Omar Iskandarani, ‘Golden Hyperbolic Swirl-Angle and Tangential Swirl Speed in Swirl–String Theory (SST)’ (2025).

The Kinematic Setting of Swirl–String Theory

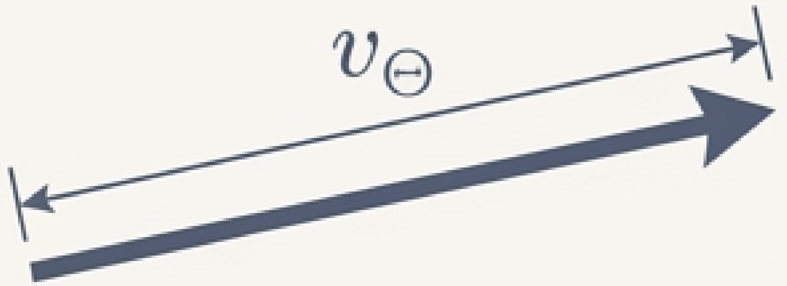
In Swirl–String Theory (SST), kinematics are described relative to fundamental constants of the theory. We focus on tangential swirl motion.

Canonical Swirl Velocity ($\boldsymbol{v}_{\ominus}$)

The theory defines a characteristic velocity vector $\boldsymbol{v}_{\ominus}$.

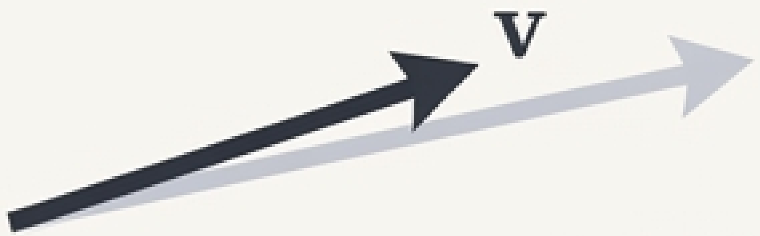
Reference Swirl Speed ($\boldsymbol{v}_{\ominus}$)

Its magnitude, $v_{\ominus} \equiv \|\mathbf{v}_{\ominus}\|$, serves as a fixed reference scale.



Local Tangential Velocity ($\boldsymbol{v}$)

The velocity of a local element, such as a swirl-string segment.



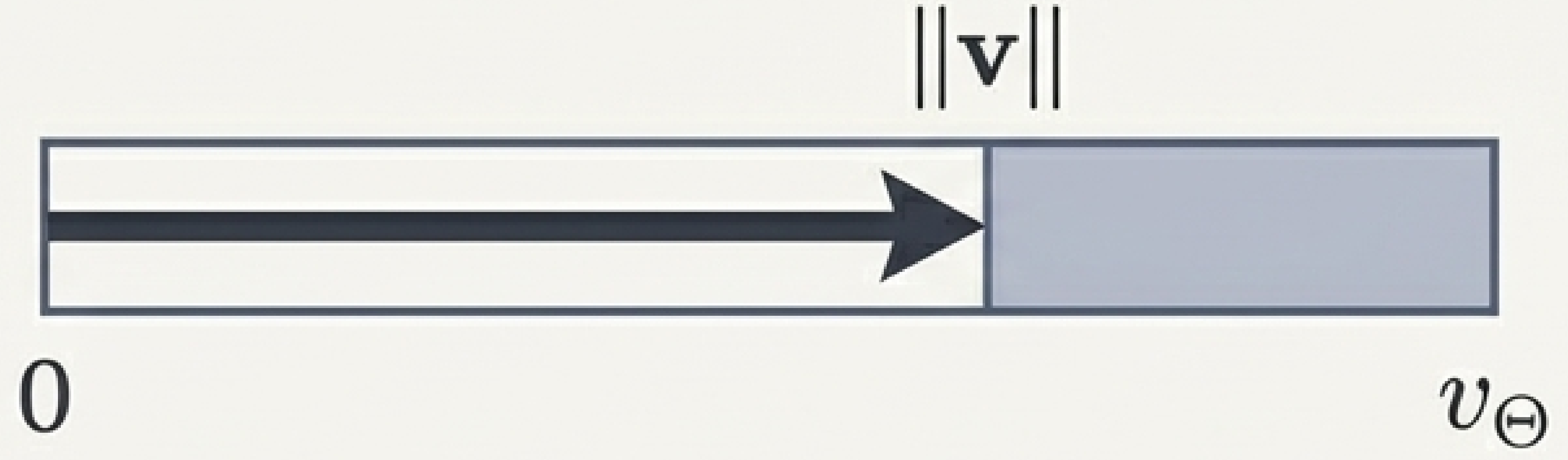
The core idea is to measure local swirl speed as a fraction of this canonical reference speed.

Parameterizing Tangential Motion: The Swirl Fraction β

The dimensionless tangential swirl fraction, β, is defined as the ratio of local swirl speed to the reference speed:

$$\beta \equiv \frac{||\mathbf{v}||}{v_\Theta}$$

By definition, β is constrained to the open interval $[0, 1)$. The reference speed v_Θ acts as the sector's upper bound.



The Challenge: Working directly with a variable bounded in an open interval can be cumbersome. A coordinate system that naturally respects this boundary is advantageous. This motivates the search for a more convenient parameterization.

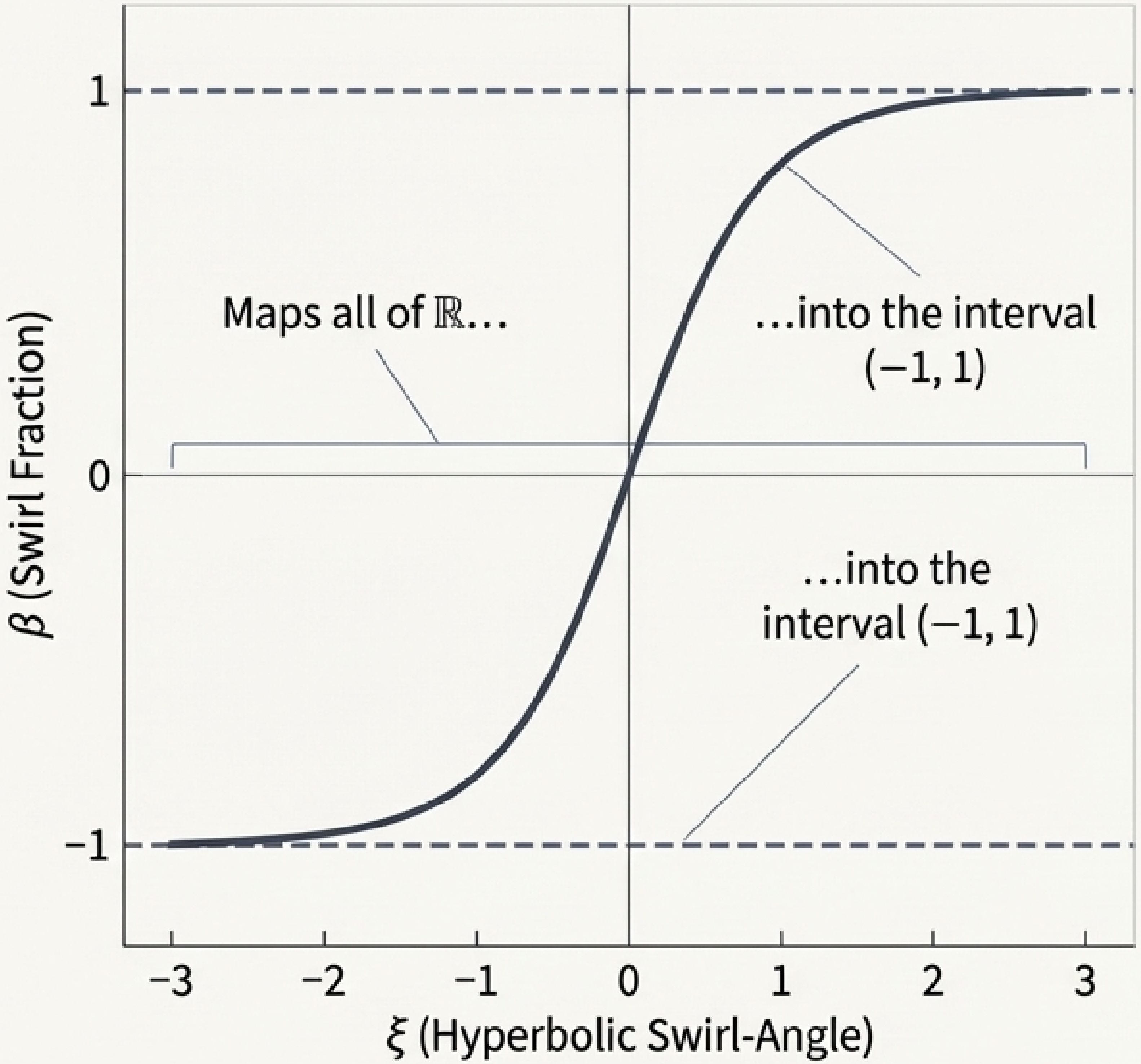
The Solution: A Hyperbolic Swirl-Angle Coordinate

We introduce a hyperbolic swirl-angle, ξ, through a bijective mapping with β.

$$\beta = \tanh \xi$$
$$\xi = \operatorname{artanh}(\beta)$$

Why This is a Powerful Choice

1. **Enforces the Bound:** The tanh function naturally maps the entire real line of ξ into the interval $(-1, 1)$, automatically satisfying $\beta < 1$.
2. **Linearizes Algebra:** It simplifies manipulations by leveraging exponential identities inherent in hyperbolic functions.
3. **Intuitive Limits:** For small speeds $(\beta \to 0)$, $\xi \approx \beta$. As speed approaches the limit $(\beta \to 1)$, $\xi \to +\infty$.



A Second Ingredient: The Golden Ratio from a Hyperbolic View

We adopt a definition of the golden ratio, ϕ, rooted entirely in hyperbolic functions.

$$\phi \equiv e^{\operatorname{asinh}(0.5)}$$

This definition relies on the standard logarithmic form of the inverse hyperbolic sine:

$$\operatorname{asinh}(x) = \ln(x + \sqrt{x^2 + 1}))$$

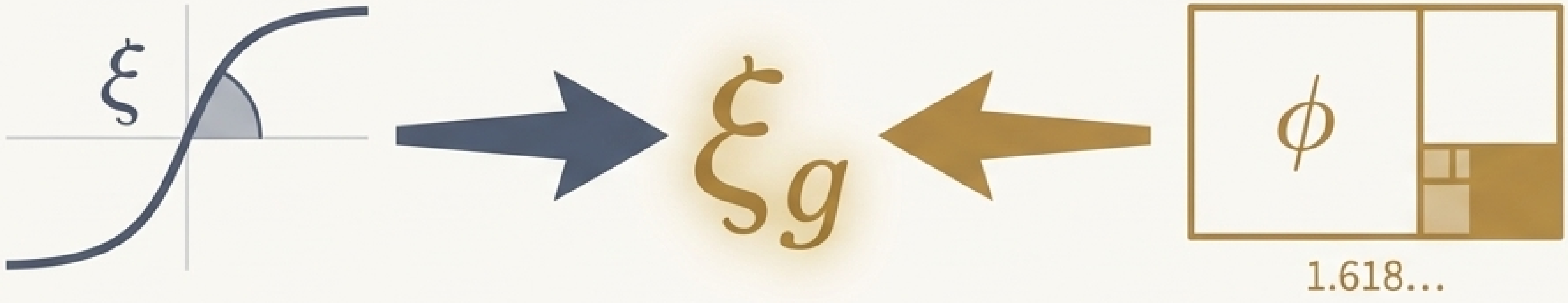
This definition may seem unfamiliar, but we will now show that it precisely recovers the well-known algebraic properties of ϕ.

Connecting to the Classics: Proof of the Algebraic Identity

Lemma 1. The hyperbolic definition $\phi \equiv e^{\operatorname{asinh}(0.5)}$ implies the familiar relation $\phi^2 = \phi + 1$.

1. Let $t := e^{\operatorname{asinh}(0.5)}$. By definition, $\phi = t$.
2. From the definition of asinh, we have $\sinh(\operatorname{asinh}(0.5)) = 0.5$.
3. Using the exponential identity $\sinh(y) = \dfrac{e^y - e^{-y}}{2}$, we substitute $y = \operatorname{asinh}(0.5)$:
$$\frac{t - t^{-1}}{2} = 0.5$$
4. Algebraic simplification leads directly to the quadratic equation:
$$t - \frac{1}{t} = 1 \Rightarrow t^2 - 1 = t \Rightarrow t^2 = t + 1.$$
5. Since $\phi = t$, we have $\phi^2 = \phi + 1$. The positive solution to this quadratic is the well-known value $\phi = (1 + \sqrt{5})/2$.

The Synthesis: Defining the Golden Hyperbolic Swirl-Angle ξ_g



We now define a special, or "golden," value for the hyperbolic swirl-angle.

The golden hyperbolic swirl-angle, ξ_g, is defined with a specific scaling of the term from our definition of ϕ.

$$\xi_g \equiv \frac{3}{2}\operatorname{asinh}(0.5)$$

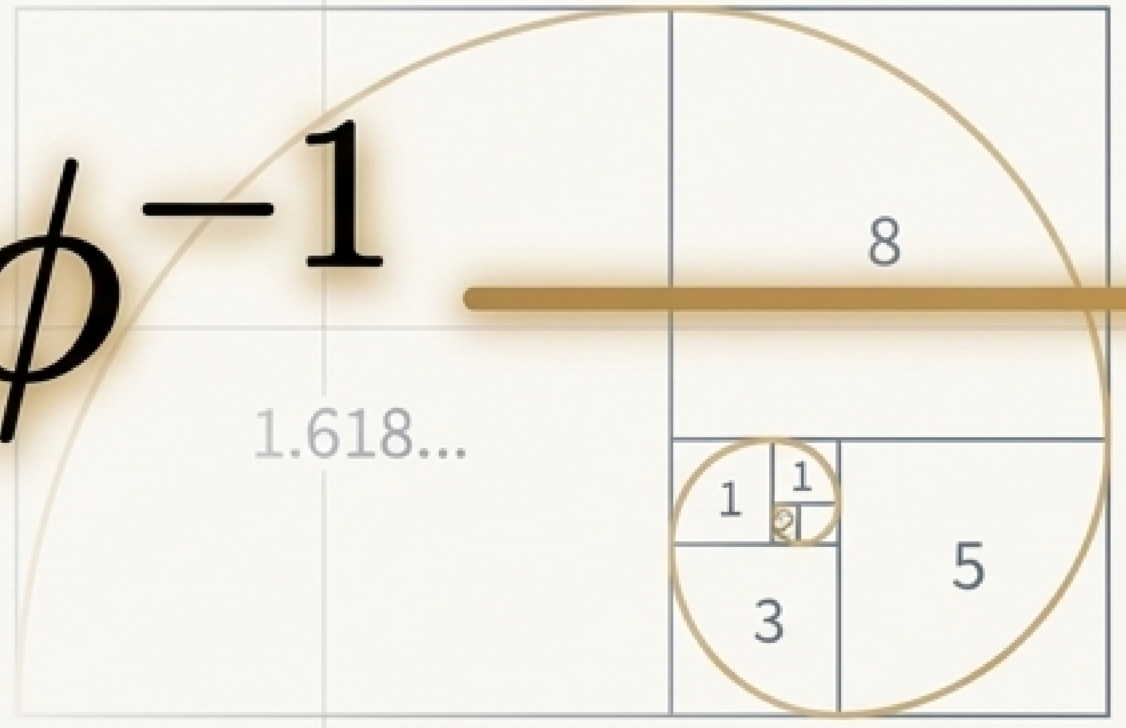
"This specific value is not arbitrary. We will now demonstrate that this precise angle corresponds to a profound physical and mathematical identity within the SST framework."

The Central Result: A Golden Connection

Theorem 1: The golden swirl-angle implies a golden tangential fraction.

At the precise angle $\xi = \xi_g$, the tangential swirl fraction β is exactly equal to the inverse of the golden ratio, ϕ^{-1}.

$$\tanh(\xi_g) = \phi^{-1}$$



Proof of the Golden Connection

To prove $\tanh(\xi g) = \phi^{-1}$.

1. Start with the identity: $\tanh(y) = \dfrac{e^{2y} - 1}{e^{2y} + 1}$.
2. Substitute $y = \xi g = \frac{3}{2}\operatorname{asinh}(0.5)$. Let $t = e^{\operatorname{asinh}(0.5)} = \phi$.
3. The expression becomes: $\tanh(\xi g) = \dfrac{e^{3\operatorname{asinh}(0.5)} - 1}{e^{3\operatorname{asinh}(0.5)} + 1} = \dfrac{t^3 - 1}{t^3 + 1}$.
4. From Lemma 1, we know $t^2 = t + 1$. We can use this to simplify t^3:
$$t^3 = t \cdot t^2 = t(t+1) = t^2 + t = (t+1) + t = 2t + 1$$
5. Substitute this back into the expression for $\tanh(\xi g)$:
$$\tanh(\xi g) = \frac{(2t+1) - 1}{(2t+1) + 1} = \frac{2t}{2t+2} = \frac{t}{t+1}$$
6. Finally, substitute t^2 for $t + 1$:
$$\tanh(\xi g) = \frac{t}{t^2} = \frac{1}{t}.$$
7. Since $t = \phi$, we conclude: $\tanh(\xi g) = \dfrac{1}{\phi} = \phi^{-1}$. □ Q.E.D.

The Implication: Defining the SST "Golden Layer"

Concept: The 'golden layer" is the kinematic state in SST defined by setting the swirl-angle to its golden value.

Definition: The golden layer occurs when $\xi = \xi_g$.

Fixed Physical Properties:

- **Golden Swirl Fraction (β_g):** At this layer, the tangential fraction is fixed:

$$\beta_g \equiv \tanh(\xi_g) = \phi^{-1}$$

- **Golden Tangential Speed ($\|\mathbf{v}\|_g$):** This implies the local tangential speed is a fixed fraction of the canonical speed:

$$\|\mathbf{v}\|_g = \beta_g * v_{\ominus} = \frac{v_{\ominus}}{\phi}$$

Further Consequences: Angular Frequency at the Golden Layer

Using the SST core length scale, r_c, we can define a corresponding angular frequency. This is a dimensional normalization, not a new dynamical assertion.

Canonical Angular Frequency (Ω)

$$\Omega \equiv \frac{v_{\circleddash}}{r_c}$$

Golden Layer Angular Frequency (Ω_g)

$$\Omega_g \equiv \frac{\|\mathbf{v}\|_g}{r_c} = \frac{1}{\phi}\frac{v_{\circleddash}}{r_c}$$

$$\boxed{\Omega_g = \frac{\Omega}{\phi}}$$

The golden layer defines not only a characteristic speed but also a characteristic frequency, both scaled by the golden ratio relative to the theory's canonical constants.

Grounding the Theory: Numerical Evaluation

Concept: We can calculate the explicit values for the golden layer properties using the canonical constants specified in SST.

Canonical Reference Swirl Speed

$$v_{\Theta} = 1.093\,845\,63 \times 10^{6}\,\mathrm{m\,s^{-1}}$$

Core Length Scale

$$r_c = 1.408\,970\,17 \times 10^{-15}\,\mathrm{m}$$

From O. Iskandarani, 2025, SST Canon v0.6.0

Next Step

Using these values, we will now evaluate ϕ, ξ_g, β_g, and the resulting physical scales $\|\mathbf{v}\|_g$ and Ω_g.

The Golden Layer: By the Numbers

Parameter	Definition	Numerical Value
Golden Ratio (ϕ)	$e^{\operatorname{asinh}(0.5)}$	≈ 1.6180339887
Intermediate Value	$\operatorname{asinh}(0.5)$	≈ 0.4812118251
Golden Swirl-Angle (ξ_{g})	$\frac{3}{2}\operatorname{asinh}(0.5)$	≈ 0.7218177376
Golden Fraction (β_{g})	$\tanh(\xi_{\mathrm{g}}) = \phi^{-1}$	≈ 0.6180339887
Golden Speed ($\lVert v \rVert_{\mathrm{g}}$)	$\frac{v_\infty}{\phi}$	$\approx 6.760\,337\,78 \times 10^{5}\ \mathrm{m\,s^{-1}}$
Canonical Frequency (Ω)	$\frac{v_\infty}{r_c}$	$\approx 7.763\,440\,66 \times 10^{20}\ \mathrm{s^{-1}}$
Golden Frequency (Ω_{g})	$\frac{\Omega}{\phi}$	$\approx 4.798\,070\,19 \times 10^{20}\ \mathrm{s^{-1}}$

What is the Contribution? An Interpretive Framework

Existing Tools

The hyperbolic function identities and the properties of ϕ are classical and well-established mathematics.

The SST-Specific Contribution is the ‘Packaging’

ξ

1. Introducing ξ as a *hyperbolic swirl-angle* that coordinates the SST tangential fraction β.

ξ_g

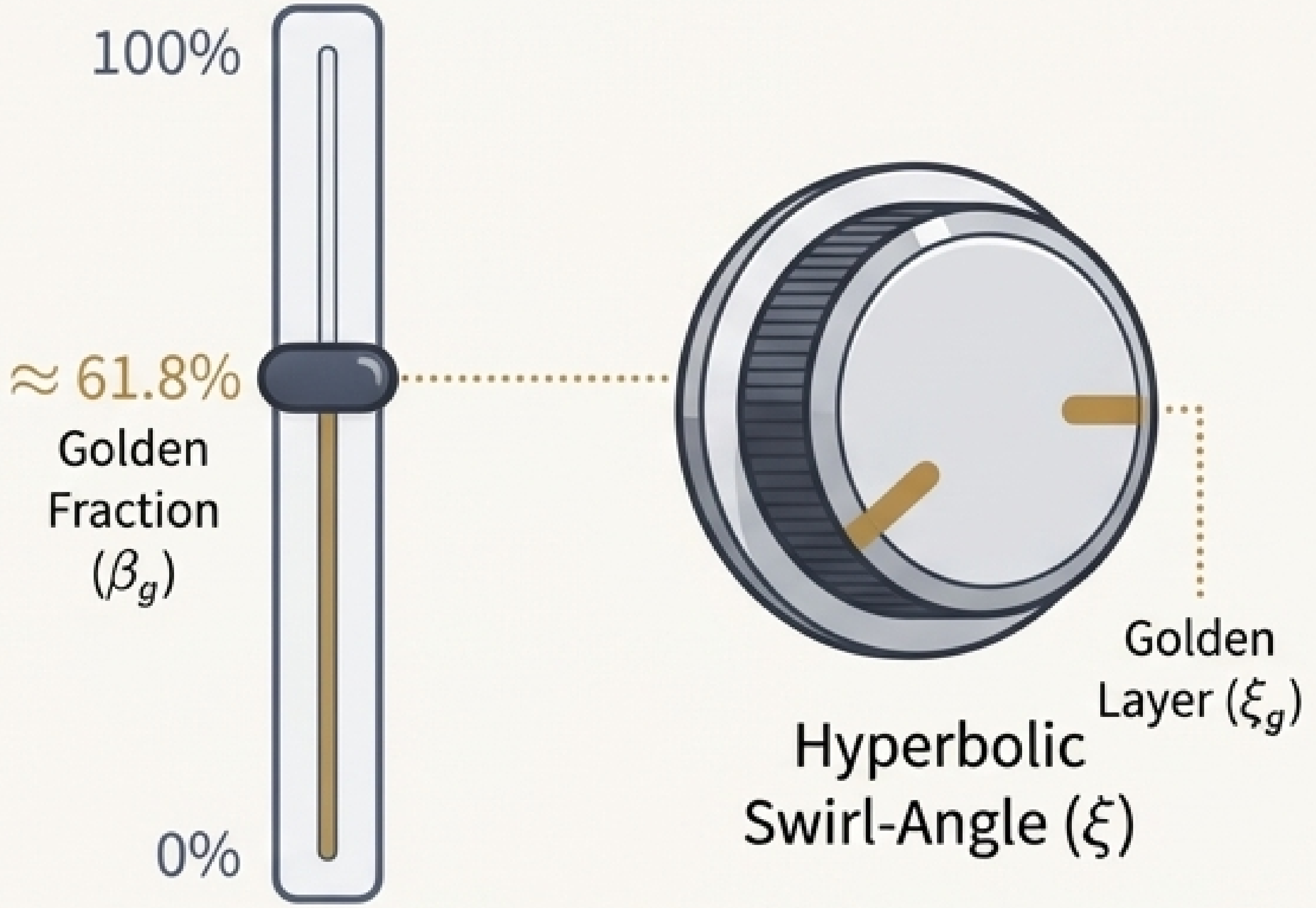
2. Defining a specific *golden layer* at ξ_g creates a dimensionless, theory-internal marker where $\beta_g = \phi^{-1}$.

3. The final step is to express the associated speed ($\|v\|_g$) and frequency (Ω_g) scales in terms of SST’s own canonical constants, v_∞ and r_c.

The novelty lies in the synthesis and application, creating a new, elegant structure within Swirl–String Theory.

An Intuitive Analogy

Imagine a speed slider that can never go above 100%. This is the swirl fraction, β.



The hyperbolic swirl-angle (ξ) is like a special knob that controls the slider. Turning this knob by equal steps moves the slider smoothly, and you can turn the knob infinitely far without ever breaking the 100% limit.

The golden layer (ξ_g) is one special, marked position on that knob. When you turn the knob **to this exact spot,** the slider **always** lands precisely at the golden fraction—about 61.8%.